

Project name: *Regulatory Reporting – Sports Direct*

Objective:

The primary goal of this project was to aggregate and analyze transaction data for Sports Direct in order to meet regulatory reporting requirements for two distinct entities: R1 (representing the UK operations) and R2 (representing the US operations). This involved consolidating customer and transaction information, ensuring data accuracy through cleaning and validation, and producing actionable insights for reporting purposes.

Focus:

The analysis specifically concentrated on distinguishing between cross-currency and same-currency transactions denominated in GBP over the period from **April 1, 2022, to August 1, 2023**. Cross-currency transactions are those where the start and end currencies differ, while same-currency transactions involve no currency conversion. The study aimed to provide both absolute monetary values and relative proportions, enabling a clear understanding of transaction volumes and trends for regulatory reporting.

Data Sources

The analysis leveraged two primary datasets—**Customers** and **Transactions**—to build a comprehensive view of Sports Direct's transaction activity across the UK and US entities.

1. Customers Table

This table contains detailed information about individual customers, including:

- **customer_id**: A unique identifier for each customer.
- **customer_type**: Classification of the customer (e.g., retail, corporate, VIP).
- **current_address_country**: The country of residence for the customer, used to determine entity affiliation (UK or US).
- **customer_since_date**: The date when the customer first joined or transacted, providing historical context for customer activity.

2. Transactions Table

This table captures transactional data for each customer, including:

- **transaction_id**: A unique identifier for each transaction.
- **customer_id**: Links the transaction to the corresponding customer in the Customers table.
- **amount_GBP**: The transaction value in GBP, serving as the primary metric for analysis.
- **currency_route**: Indicates the start and end currency for the transaction, enabling differentiation between cross-currency and same-currency transactions.
- **transaction_date**: The date the transaction occurred, which was filtered for the period **01/04/2022 – 01/08/2023** for the reporting scope.

Relationship Between Tables

The two tables were joined on **customer_id** to create a consolidated dataset. This allowed each transaction to be enriched with customer attributes, enabling entity-specific analysis and more accurate aggregation of cross-currency and same-currency GBP volumes.

Visual Suggestion: A simple diagram showing Customers → Transactions relationship, with arrows linking **customer_id** as the key join field.

Data Cleaning & Preparation

- **Removed duplicates:** Ensure each `transaction_id` is unique.
- **Excluded negative values:** Focus on positive GBP transactions only.
- **Standardized text fields:** Trimmed whitespaces in `customer_type` and `currency_route`.
- **Joined tables:** Merged Customers & Transactions on `customer_id` → created `customer_transactions` dataset.
- **Result:** Clean, consolidated dataset ready for analysis and visualization.
- **Purpose:** Ensures accurate aggregation of metrics for R1 (UK) and R2 (US) entities.
- **Next step:** Use this dataset to calculate cross-currency vs same-currency transaction volumes.

Visual Suggestion: Simple flow diagram:

Raw Customers & Transactions → Cleaning → Join → Customer_Transactions Dataset

Metrics Calculation

● Cross-Currency vs Same-Currency:

- *Cross-Currency*: Start and end currencies differ.
- *Same-Currency*: Start and end currencies are the same.
- **R1 – UK Entity**: Filter by `current_address_country = 'UK'` and transactions including GBP.
- **R2 – US Entity**: Filter by `current_address_country = 'USA'` or `transaction_origin_country = 'USA'`.

● Metric Types:

- *Absolute values*: Total GBP volume per category.
- *Relative values*: Percentage of total GBP volume.
- **Visuals**: Bar and pie charts to compare cross vs same-currency transactions.

R1 – UK Entity Metrics

- **Filters applied:**

- `current_address_country = 'UK'`
- Transactions including GBP
- Date range: 01/04/2022 – 01/08/2023

- **Metrics:**

- *Cross-Currency GBP*: Total value of transactions with currency conversion
- *Same-Currency GBP*: Total value of transactions without conversion
- *Relative values*: Percentage of total GBP volume

- **Visuals:**

- Bar chart for absolute GBP values
- Pie chart for relative proportions

- **Insight:** Quickly compare cross-currency vs same-currency activity for the UK entity.

R1 - Relative vs Absolute



Measure Names

- Absolute Value Gbp
- Relative Value Gbp

R2 – US Entity Metrics

- **Filters applied:**

- `current_address_country = 'USA'` OR `transaction_origin_country = 'USA'`
- Date range: 01/04/2022 – 01/08/2023

- **Metrics:**

- *Cross-Currency GBP*: Total value of transactions with currency conversion
- *Same-Currency GBP*: Total value of transactions without conversion
- *Relative values*: Percentage of total GBP volume

- **Visuals:**

- Bar chart for absolute GBP values
- Pie chart for relative proportions

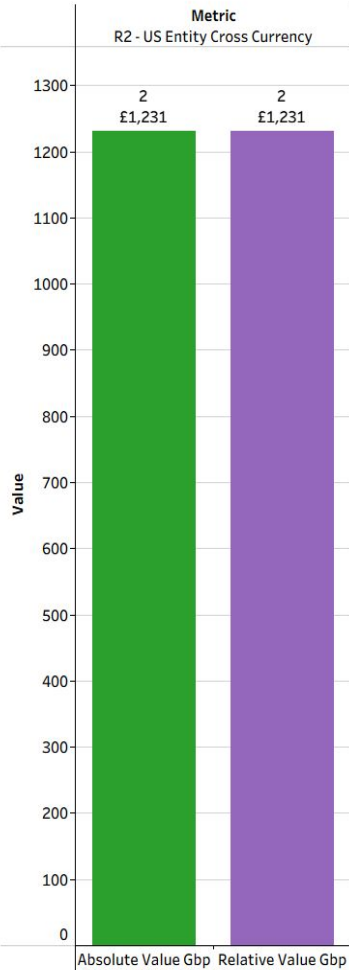
- **Insight:** Compare cross-currency vs same-currency activity for the US entity and identify trends relative to the UK entity.

R2 US Cross-Currency

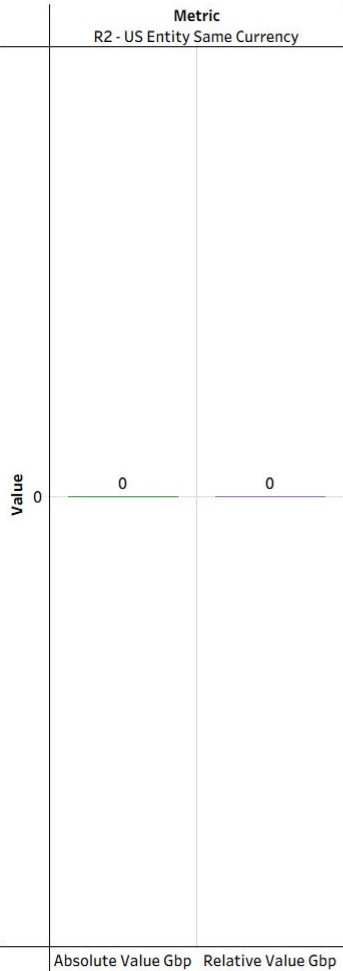
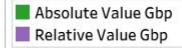
Measure Names

Absolute Value Gbp

Relative Value Gbp



R2 US Same-Currency



Some categories show zero values because there were no transactions recorded for that currency type or entity during the selected period.

Absolute vs Relative Values & Assumptions / Limitations

- **Absolute values:** Represent total GBP transaction amounts for each category.
- **Relative values:** Show the proportion of each category, highlighting trends and comparisons.
- **Assumptions:**
 - UK and US entities defined by country fields
 - Cross/same-currency based on `currency_route`
 - Transactions limited to 01/04/2022 – 01/08/2023
- **Limitations:**
 - Some transactions missing `currency_route`
 - Negative amounts excluded (refunds/reversals not included)
 - Exchange rates not applied; relative values may exaggerate small categories

Visual suggestion: Example chart showing absolute vs relative comparison

SQL Logic & Methodology

- **Data Extraction:** Imported Customers & Transactions CSVs into BigQuery.
- **Data Cleaning:** Removed duplicates, negative values, trimmed text fields.
- **Table Join:** Merged Customers & Transactions on `customer_id` → `customer_transactions` dataset.
- **Metrics Calculation:**
 - Identified **cross-currency** vs **same-currency** transactions
 - Aggregated **absolute** and **relative** GBP values for R1 (UK) & R2 (US)
- **Validation:** Checked for missing values, duplicates, and zero-value transactions.
- **Visualization:** Prepared charts in Tableau to compare metrics across entities.

Visual suggestion: Flow diagram showing SQL steps → Metrics → Charts

Example SQL snippet used to calculate UK cross-currency GBP volume (R1):

Untitled query

Run Save Download Share Schedule Open in More

```
17 ---Cross-Currency vs Same-Currency Volume
18 ---UK Entity Cross-Currency Volume
19 SELECT SUM(amount_GBP) AS cross_currency_gbp_uk
20 FROM `clientprojects-469115.ecommerce_analysis.customer_transactions`
21 WHERE transaction_date BETWEEN '2022-04-01' AND '2023-08-01'
22     AND current_address_country = 'UK'
23     AND currency_route LIKE '%GBP%'
24     AND SPLIT(currency_route, '->')[OFFSET(0)] != SPLIT(currency_route, '->')[OFFSET(1)];
25
26
27
28
29
30
31
32
```

✓ This script will process 17.34 KB when run.

Query results

Save results Open in

Job informationResultsVisualizationJSONExecution detailsExecution graph

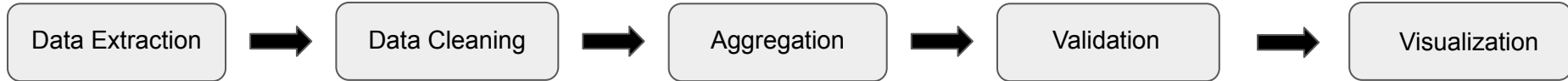
Row	cross_currency_g...
1	4320

Results per page: 50 1 – 1 of 1 < >

Regulatory Reporting Workflow & Tools

Workflow Steps:

1. **Data Extraction:** Import Customers & Transactions CSVs into BigQuery
2. **Data Cleaning:** Remove duplicates, negative values, trim text fields
3. **Aggregation:** Join tables & calculate metrics (cross vs same-currency)
4. **Validation:** Check for missing values, duplicates, zero transactions
5. **Visualization:** Prepare charts in Tableau
6. **Slide Pack Creation & Submission:** Summarize findings and share with stakeholders



Tools & Technologies: BigQuery (data storage & querying), SQL & Excel/Sheets (data cleaning & preparation), Tableau Desktop/Public (visualization), Google Slides/PowerPoint/PDF (reporting & slide pack)

Key Insights & Conclusion

- **Key Insights:**

- R1 (UK) vs R2 (US) show different levels of cross-currency activity
- Relative values highlight clear trends over time
- Zero or missing categories indicate limited or no activity in some segments

- **Conclusion / Next Steps:**

- Cleaned and aggregated data enabled accurate reporting for both entities
- Absolute & relative views provided both scale and proportion insights
- Future improvements: include refunds/chargebacks, factor in exchange rates, and automate dashboards for ongoing monitoring